FH JOANNEUM
Electronic Engineering

# Project Management

Syllabus

DI (FH) DI Gerald Derflinger, PMP

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# 1 COURSE DESCRIPTION

## 1.1 Scope and curriculum location

The course Project Management (Seminar) takes place in the 3rd semester of the Master's degree programme Electronic Engineering and has the following scope:

- 2 teaching units per week (30 teaching units in total)
- 3 ECTS credits

Course leader: DI (FH) DI Gerald Derflinger, PMP

## 1.2 Content

The course gives an overview of classic and agile project management. Students learn how to define, plan, implement, control and close a project, based on the IPMA and PMI standards and on Scrum. Topics: project types and models, project roles, project definition (objectives, requirements, organisation), project planning (stakeholder analysis, WBS, work packages, effort estimation, schedule, resources, costs), risk management and (D)FMEA, project controlling and change management, project closing, agile project management, Scrum, hybrid approaches and agile leadership principles. In a group project, students plan a hybrid hardware/software development project and present their project handbook.

## 1.3 Previous Knowledge Expected

Students are expected to have prior knowledge in the following areas:

- MS Office, especially Excel (or MS Project) for scheduling

## 1.4 Objective

The students

- know the characteristics of projects, the project phases and the main project management standards (IPMA, PMI, Scrum),
- can define a project: project charter, SMART objectives, stakeholder and environment analysis, project organisation,
- can plan a project: work breakdown structure, work packages, responsibility matrix (RACI), schedule incl. critical path, resource and cost plan,
- can identify, assess and mitigate project risks and know the basics of (D)FMEA,
- know agile values, principles and Scrum and can decide when to use a plan-driven, an agile or a hybrid approach,
- can control and close a project (milestone trend analysis, change management, lessons learned) and know principles of agile leadership.

## 1.5 Teaching and Learning Methods

The course content is taught in

- lectures with examples from industrial practice,

- project work in groups (demo project),
- presentations of the project results.

The course is consolidated by means of a group project in which students plan a complete development project and present it.

## 2 INFORMATION ON THE EXAMINATION

### 2.1 Assessment

The following performances are used for the assessment of the course:

- individual written examination (test, 60 min.)
- project work in groups: project handbook, project schedule and presentation
- active participation in the course

As part of the performance assessment, an oral clarification discussion may take place to determine the independence of the work performed. Students are obliged to take part in this.

### 2.2 Scoring

The final grade is made up of the results of the partial performances listed above as follows:

- written examination: 50 points
- project work (project handbook, project schedule, presentation): 30 points
- active participation: 20 points
    - attendance of at least 80 %: 10 points (below 80 %: 0 points)
    - active contribution to the group project: up to 10 points

The individual points are not taken into account as grades but sum up to a total score of max. 100 points.

### 2.3 Grading

The final mark is defined by the total score, using the following scheme:

- 90 – 100 Very Good (1)
- 80 – 89,9 Good (2)
- 65 – 79,9 Satisfactory (3)
- 50 – 64,9 Sufficient (4)
- 0 – 49,9 Not Sufficient, failed (5)

Prerequisites for a positive final grade are: at least 50 % of the points of the written examination (25 points), at least 50 % of the points of the project work (15 points), the project handbook and the project schedule have been submitted in time, and attendance of at least 80 %.

### 2.4 Allowed resources for written examinations

The following aids and resources are permitted in the written examinations:

- calculator without computer algebra system

Please note that only the aids listed above are permitted.

### 2.5 Use of generative AI tools

Generative AI tools may be used in this course for specific tasks. Permitted is the use for:

- literature research and understanding of concepts,
- language correction and translation of texts you wrote yourself.

Not permitted is the use for:

- generating entire text sections of the project handbook or the presentation,
- creating the WBS, work package specifications, schedule, effort and cost estimates and the risk assessment of the project work,
- the written examination (no electronic devices except the calculator).

In any case, the following rules apply:

- Please indicate when you use generative AI tools for your work. A declaration of independence, which also contains information about the AI used and how it was used, is required for every submitted work.
- Document every AI tool used with its exact name and version, provider, date of use, URL and purpose. AI-generated content is not a scientific source; AI-generated passages must be marked and cited.
- Please note that you bear full responsibility for the accuracy of the AI-generated content.
- Contributions that have been created predominantly or even exclusively using generative AI are considered 'fraudulent' in accordance with Section 17 (8) of the StuPO and declared invalid.

### 2.6 Repetition Options

The first examination is offered on two dates: 1a on Dec. 16th 2026 and 1b (date will be announced). In the case of a negative assessment, the following partial performances are to be performed again:

- written examination, if less than 25 points were achieved,
- project work, if less than 15 points were achieved: the project handbook has to be revised and resubmitted within a deadline communicated in writing.

As the group work cannot be repeated in the following semester, a substitute performance (individual revision of the project handbook) may be required.

The board repetition of an examination is always an overall examination. However, thematic focal points can be set.

Examinations that have been passed can be repeated once. The repetition of a positively completed examination must be reported to the head of the degree programme within two weeks of the announcement of the examination result. If several partial performances are to be achieved, the course instructor determines which ones are to be repeated. Registration for the examination is required for the repetition. This repetition counts towards the total number of examination attempts.

For each retake exam, only the grade of the retake exam counts.

### 2.7 Attendance requirements

Students must attend at least 80% of the courses.

## 3 ADDITIONAL INFORMATION

### 3.1 Recommended reading

[1] Project Management Institute: A Guide to the Project Management Body of Knowledge (PMBOK® Guide), current edition.

[2] IPMA: Individual Competence Baseline (ICB), Version 4.0.

[3] Ken Schwaber, Jeff Sutherland: The Scrum Guide, 2020. https://scrumguides.org

[4] Jurgen Appelo: Management 3.0 – Leading Agile Developers, Developing Agile Leaders. Addison-Wesley.

[5] Mitch Lacey: The Scrum Field Guide – Agile Advice for Your First Year and Beyond. Addison-Wesley.

### 3.2 Hardware and Software

MS Word for the project handbook; MS Excel, MS Project or a comparable tool for the project schedule.

### 3.3 Contents of the individual teaching units

The following table lists the type and scope of the planned teaching units and their contents. The actual sequence of the teaching units can be found in the timetable.

| Unit | Type/Scope | Content |
|---|---|---|
| **1** | 4 LE | Introduction; project management basics, project types and models; demo project and group formation |
| **2** | 4 LE | Project roles; project definition: objectives, requirements, project organisation, team development |
| **3** | 4 LE | Project planning I: stakeholder and environment analysis, work breakdown structure, work packages, effort estimation, communication plan |
| **4** | 4 LE | Agile project management; introduction to Scrum |
| **5** | 4 LE | Project planning II: schedule, resources, costs, risk management and (D)FMEA; project implementation and controlling |
| **6** | 4 LE | Project closing; agile leadership principles; exam preparation |
| **7** | 2 EXM | Written examination (date 1a) |
| **8** | 4 LE | Presentations of the group projects (15 min. + 5 min. questions per group) |

- LE Lecture
- EXE Exercise Computer
- EXL Exercise Laboratory
- EXM Exam